

SonarQube Installation on AWS EC2 (RHEL/Amazon Linux) using Java 21 (Amazon Corretto)

Brief, production-ready steps to install SonarQube 10.4.x on EC2 with Amazon Corretto 21 and required kernel settings.

1 Prerequisites

- EC2: RHEL 8/9 or Amazon Linux, 4+ GB RAM
- Security Group: TCP 9000 open
- User: ec2-user (do not run as root)
- Update OS:

```
sudo dnf -y update
```

2 Install Java 21 (Amazon Corretto)

```
sudo dnf -y install java-21-amazon-corretto-devel
```

or

```
sudo yum -y install java-21-amazon-corretto-devel
```

verify

```
java -version
```

Reason: SonarQube 10.4.x supports Java 21.

3 Configure JAVA_HOME

```
echo 'export JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto' >> ~/.bashrc
```

```
echo 'export PATH=$JAVA_HOME/bin:$PATH' >> ~/.bashrc
```

```
source ~/.bashrc
```

4 Download SonarQube

```
sudo dnf -y install unzip
```

```
wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.4.1.88267.zip
```

```
unzip sonarqube-10.4.1.88267.zip
```

5 Set Permissions (Important)

```
sudo chown -R ec2-user:ec2-user /home/ec2-user/sonarqube
```

SonarQube must not run as root.

6 Kernel Configuration (Elasticsearch)

```
echo 'vm.max_map_count=262144' | sudo tee -a /etc/sysctl.conf
```

7 Configure sonar.properties (Java 21 Fix)

Edit /home/ec2-user/sonarqube/conf/sonar.properties and add at the last line:

```
sonar.web.javaOpts = -Xms512m -Xmx512m -Djava.security.manager=allow
```

```
sonar.ce.javaOpts = -Xms512m -Xmx512m -Djava.security.manager=allow
```

```
sonar.search.javaOpts = -Xms1g -Xmx1g -Djava.security.manager=allow
```

⚠ Java 21 requires -Djava.security.manager=allow. Do not use deprecated properties.

8 Clean Temp Directory

```
rm -rf /home/ec2-user/sonarqube/temp/*
```

9 Start SonarQube

```
cd /home/ec2-user/sonarqube/bin/linux-x86-64
```

```
./sonar.sh start
```

```
./sonar.sh status
```

if it fails running or after sometime if it crashes check the logs at

```
tail -f ../../logs/sonar.log
```

10 Access SonarQube UI

- URL: http://publicip:9000
- Default login: admin / admin

⚠ Common Failures & Fixes

- Crash after startup → Ensure javaOpts include -Djava.security.manager=allow in all three sections.
- Elasticsearch exit 143 → Increase vm.max_map_count, ensure 4+ GB RAM, adjust search heap if needed.

- Permission errors → Re-run chown; ensure you start as ec2-user.
- Java mismatch → Verify JAVA_HOME and java -version show Corretto 21.